

Ex.No.4 VERIFICATION OF GPIO PORTS OF 8051

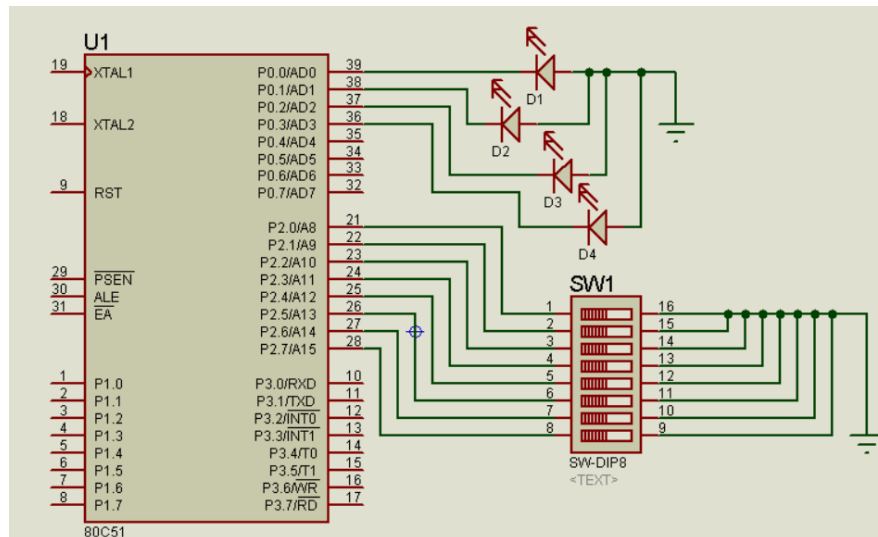
AIM

To verify GPIO ports of 8051 using embedded C programming.

EQUIPMENT

Keil uvision-IDE, 8051 development board, FRC cables

CIRCUIT DIAGRAM



PROCEDURE

Step 1: Give a double click on µvision 5 icon.

Step 2: To create a new project, click on the project window and select new µvision project. Create a new folder with a project name to save the project.

Step 3: Select the target microcontroller i.e. 8051AH from Intel and click OK

Step 4: Click YES to Copy STARTUP.A51 to Project Folder and Add file to Project. Type the Embedded C Program and save the file in the folder created with c extension.

Step 5: Add the .c File to source group by right click on Source Group, choose “**ADD files to Group**” option.

Step 6: Right click on target in the project window and select “**Options for Target**”

Step 7: Go to “**OUTPUT**” option in the popped window and check (✓) “**Create Hex File**”.

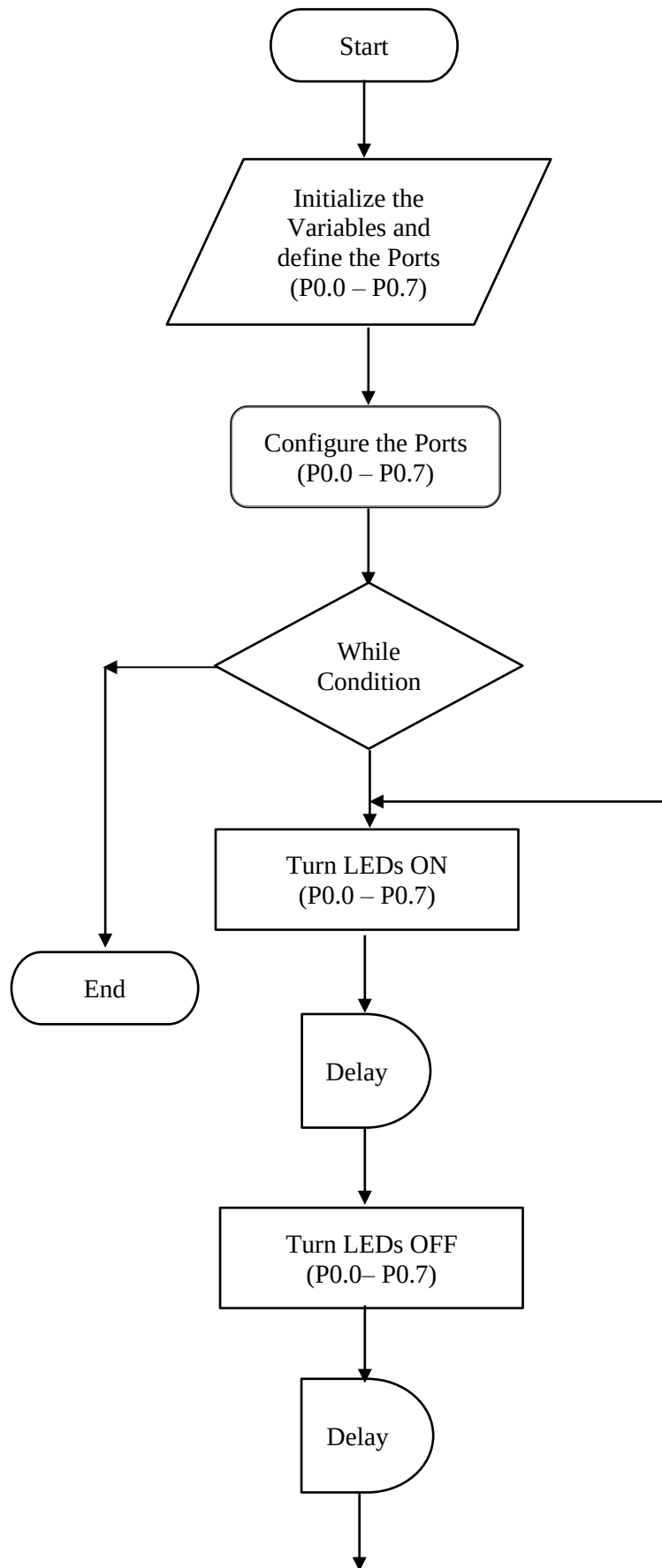
Step 8: In the same window go to Linker option and check (✓) “**Use Memory Layout from Target Dialog**” and click OK.

Step 9: To compile the project, click Project and select Build Target option or press F7.

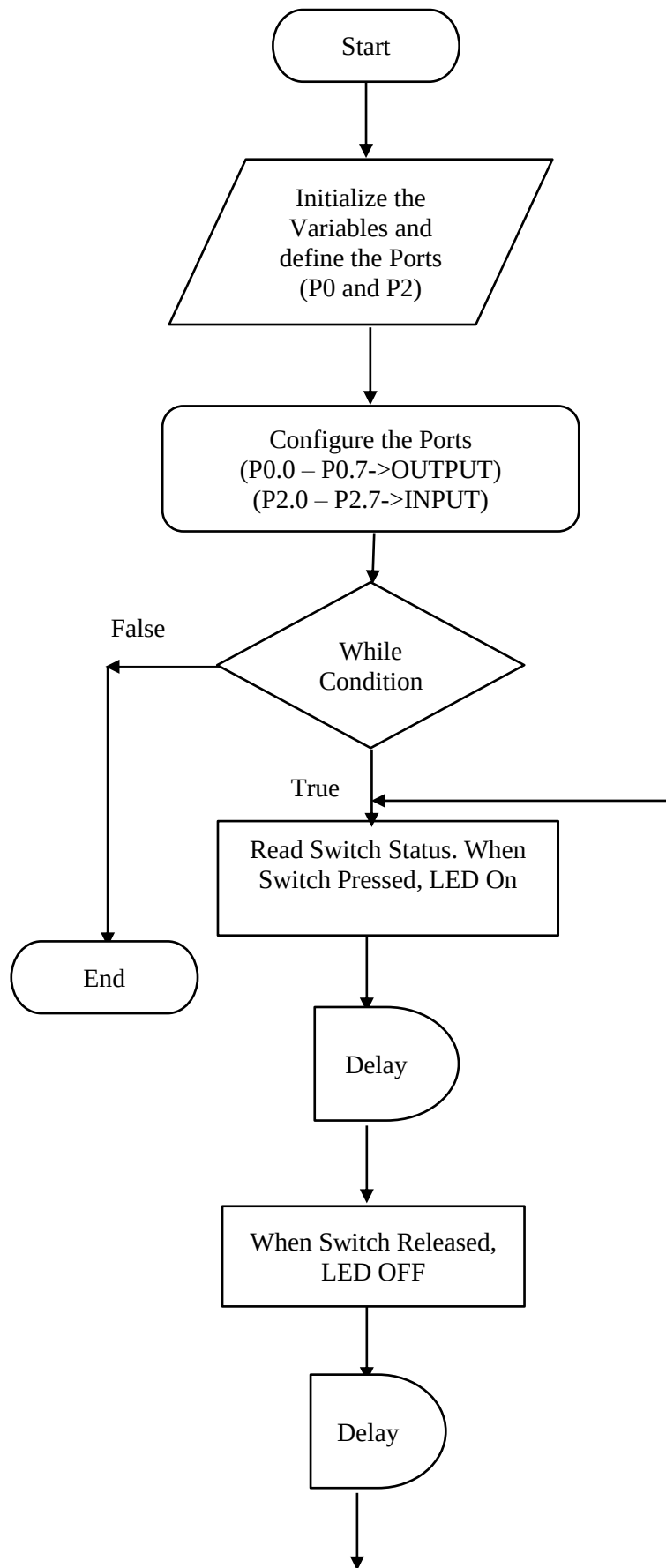
Step 10: Next open Progisp Software, select the microcontroller (AT89S52), enable “verify signature”, “Chip Erase”, “program FLASH” and “verify FLASH”. Click Load Flash to burn the code (hex file) into the controller. Click on the AUTO button to upload the code.

Step 11: Check the output of individual LEDs as individual switches are turned ON in the development board.

FLOWCHART-FLASHING LED



FLOWCHART- LED - SWITCH INTERFACE



ALGORITHM- FLASHING LED

1. Start the program.
2. Initialize the header files.
3. Declare the delay functions.
4. Declare the direction of port pins (P0.0-P0.7) as output.
5. Send High value to the port pins (P0.0 – P0.7)
6. Call the delay function.
7. Send Low value to the port pins P0.0 – P0.7)
8. Call the delay function.
9. To continue the LEDs flashing repeat from step 5.

ALGORITHM- LED- SWITCH INTERFACE

1. Initialize the header file <reg51.h>.
2. Define Port '0' as output port and Port '2' as input port.
3. Declare variable names for individual switch and individual LED to call them in the main program.
4. Monitor the status of the pins connected to the pushbutton switch.
5. Check the status of switch in P2 and infer output of LED in P0
6. Repeat the steps for all the switches and check the LED outputs.

PROGRAM

RESULT

Thus, verification of GPIO ports of 8051 using embedded C programming is done.